RESEARCH INTERNSHIP — AI & QUANTUM HARDWARE

February – July 2027 · 6 months

Physics and AI student working at the intersection of machine learning, quantum computing, and experimental quantum platforms.

YASSINE FERHI

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EXPERIENCE

Research Internship – Quantum Systems Research Group

05/2026 - 09/2026
Trieste, Italy

National Research Centre – National Institute of Optics

- Developed a gradient-descent algorithm for automated MOT parameter optimization, increasing atomic density by 1.5× and reducing daily optimization time from 2 hours to less than 1 hour.
- Designed an adaptive laser-control loop based on experimental feedback, extending laser stability from 3–4 days to several weeks.
- Automated the optimization of a cold-atom experimental platform, integrating real-time experimental measurements into the control loop.

Research Internship – Quantum Image Processing Department

06/2025 - 08/2025
Kaiserslautern, Germany

Institut Fraunhofer (ITWM)

- Designed and evaluated the performance of quantum algorithms for image processing.
- Conducted a comparative analysis of quantum pattern-matching methods, integrating a Quantum Support Vector Machine (QSVM) classifier.
- Developed an algorithm for approximating image convolution using the Quantum Fourier Transform (QFT), designed as a computational building block for artificial intelligence models.

EDUCATION

Thematic Research Year in Artificial Intelligence (ARIA)

2024 - 2028

École normale supérieure de Paris Saclay - ENS

Master's Degree (M1) in Fundamental Physics

2025 - 2026

Université Paris-Saclay - UPS

Engineering Degree in Optics and Photonics

2026 - 2027

Institut d'Optique Graduate School - IOGS

MP2I/MPI* Preparatory Classes

2022 - 2024

Lycée Hoche

SPORTS & CULTURE

Co-founder and Facilitator of the IOGS Focale Debate Club

- Founded in 2024, focusing on the ethical and societal challenges facing tomorrow's engineers.
- Organized roundtable discussions: defining topics, selecting speakers, preparing discussions, and coordinating logistics.

Kayaking



LANGUAGES

English	fluent	●●●●●
French	native	●●●●●

PUBLICATIONS

Pattern Matching Using Sliding Windows on Quantum Computing
Ferhi, Geng, Kiefer-Emmanouilidis, Moghiseh · QUEST-IS 2025 · Springer, CCIS 2744 · Publié

Approximate Image Convolution via Hadamard-Type Multiplication in the Quantum Fourier Domain
Ferhi, Geng, Kiefer-Emmanouilidis, Moghiseh · IEEE QAI 2026 · En review

Fourier Finite-Difference Method and Proper Orthogonal Decomposition for Fast Spectral Analysis of 2D Periodic Photonic Structures
Besbes, Ferhi, Benisty, Sauvan · JOSA A 43, 1380–1393 (2026) · Publié

Schmidt Decomposition-Based Methods for Efficient Quantum Image Encoding
Pangeva, Ferhi, Geng, Weinmann, Ivanova, Moghiseh · arXiv:2606.10874 (2026)

SKILLS

Programming:
- Python ●●●●●
- C ●●●●●
- OCaml ●●●●
- Quantum simulation ●●●●●

Artificial Intelligence:
- Machine Learning ●●●●
- Numerical Optimization ●●●●●
- Image Processing ●●●●●

Physics & Photonics:
- Optics & Photonics ●●●●●
- Physical Modeling ●●●●●

Soft Skills:
- Scientific rigor and analytical thinking
- Research project management
- Science communication and knowledge transfer
- Multidisciplinary teamwork
- Autonomy and organization
- Written and oral scientific communication